Marcos Esparza

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Education

Bachelor of Science: Mechanical Engineering Aug 2022 - December 2026
University of Texas Permian Basin | GPA: 3.0 *Odessa, TX*

Relevant Coursework: Thermodynamics II, Fluid Mechanics II, Mechanics of Materials, Thermo-Fluids Lab, Engineering Design
Activities: FAST, ASME, AIAA, SPE

Technical Projects

Falcon Aeronautics & Space Team (FAST) – IREC 10k COTS Rocket
Aerodynamic Design Lead – Fin Can & Structures | UTPB | Aug 2024 – Present

- Lead aerodynamic and fin-can design for FAST's 10,000-ft COTS rocket entry in the Spaceport America Cup and Lone Star Cup.
- Built SOLIDWORKS models of the airframe and fin can, integrating 98 mm motor, recovery hardware, and avionics bays.
- Performed stability and fin-flutter analysis in OpenRocket and MATLAB, maintaining a static margin of 2.88 calibers and a flutter margin >1.5× maximum flight velocity.

Thermo-Fluids / Mechanical Systems Lab – UTPB
Undergraduate Lab Engineer | Jan 2024 – Present

- Ran experiments on Pelton turbine, cross-flow heat exchanger, and flow-meter rigs to measure efficiency, head loss, and heat-transfer coefficients.
- Processed pressure/temperature/flow data in MATLAB and Excel to compare experimental results to theory and quantify error sources.
- Wrote ASME-style reports with uncertainty analysis and discussion of system performance, using NIST data and standards for reference.

L2 High-Power Rocket (Personal Project)

- Designed and built a 4″ dual-deploy cardboard airframe with fiberglass reinforcement, 38 mm motor mount, and TTW fins for NAR/Tripoli L2 certification.
- Modeled performance in OpenRocket (≈4,100 ft apogee on J425) and sized recovery system (dual-deploy, 36″ main, 14″ drogue) for safe descent rates.

Fluid Friction Experiment (H408)

- Wrote ASME-style reports with uncertainty analysis and system-performance discussion, including fluid-friction/bend-loss tests on the TecQuipment H408 (measured pressure drop, head loss, and loss coefficients vs Moody/Blasius correlations).

Experience

Logistics Associate — Academy Sports + Outdoors, Midland, TX
Jul 2024 – Present

- Process and stage online and in-store pickup orders, consistently meeting same-day deadlines.
- Perform cycle counts and resolve stock discrepancies to maintain accurate inventory.

Overnight Deli Production — H-E-B, Midland, TX
Oct 2022 – Jan 2024

- Prepared and stocked deli products overnight to support next-day store operations.
- Helped train new team members on shift procedures and safe handling practices.

Team Member — Whataburger, Midland, TX
Oct 2020 – Mar 2022

- Supported front counter and kitchen operations during high-volume periods while maintaining order accuracy and cleanliness.

Technical Skills

Programming: MATLAB, Python (basic), LabVIEW (exposure)
Software: SOLIDWORKS, OpenRocket, Excel, Word, PowerPoint
Test & Lab: Pressure/temperature instrumentation, flow meters (orifice, venturi, rotameter), Pelton turbine rig, cross-flow HX
Fabrication: Basic machining & hand tools, 3D printing, epoxy layups, high-power rocketry construction (TTW fins, dual-deploy, avionics integration)
Engineering: Aerodynamic & structural design, test data analysis, systems integration, mechanical assembly, technical writing
Certifications: Tripoli/NAR High-Power Rocketry Level 1; Level 2 in progress